University of St.Gallen

Course and Examination Fact Sheet: Autumn Semester 2025

7,082: Behavioral Science & Technology

ECTS credits: 4

Overview examination/s

(binding regulations see below)
decentral - Presentation, Analog, Group work group grade (50%)
Examination time: Term time
decentral - Presentation, Analog, Individual work individual grade (30%)
Examination time: Term time
decentral - Active participation, Analog, Individual work individual grade (20%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
7,082,1.00 Behavioral Science & Technology -- English -- de Bellis Emanuel , Stachl Clemens

Course information

Prerequisites

The course does not have any hard requirements. However, the following will be helpful:

- Basic knowledge of marketing fundamentals and consumer behavior
- General interest in empirical research and experimental designs
- Elementary statistics knowledge (descriptive statistics, inference statistics)
- Curiosity about how new technologies transform our society

Learning objectives

After taking this course, you should:

- Have developed a sophisticated understanding of how new technologies shape and are shaped by consumer behavior
- Be able to apply psychological insights to understand consumer-technology interactions and improve marketing effectiveness
- Master the analysis of behavioral data using state-of-the-art data science techniques
- Be proficient in designing behavioral experiments to establish causal relationships
- Successfully complete an empirical group project integrating different disciplinary perspectives
- Be able to communicate research findings effectively through professional presentations and written reports
- Understand the ethical implications of technology in marketing practice and implement open science standards

Content

In an era where AI personalizes every customer interaction, social robots influence purchasing decisions, and algorithms predict consumer behavior better than consumers themselves, mastering the intersection of behavioral science and technology has become essential for marketing leaders. This course equips future marketing managers with both the analytical skills and psychological insights needed to thrive in today's data-driven marketplace.

The course directly addresses critical challenges facing modern marketers: How do we design digital experiences that convert? What psychological principles drive technology adoption? How can we ethically leverage behavioral data for competitive

advantage? Through rigorous empirical methods and real-world applications, students learn to answer these questions with confidence.

This course takes an **interdisciplinary approach**, preparing marketing managers to leverage insights from multiple fields:

- **Marketing and Consumer Behavior**: Understanding consumer decision-making, brand relationships, and purchase behavior in digital contexts
- **Psychology**: Cognitive processes, emotional responses, and behavioral patterns that drive consumer actions
- **Human-Computer Interaction**: How consumers engage with digital interfaces, AI assistants, and emerging technologies
- **Data Science**: Advanced analytics for uncovering behavioral patterns and predicting consumer responses

This interdisciplinary foundation enables students to move beyond surface-level metrics to understand the psychological drivers of marketing effectiveness. Through hands-on experience with **open science tools** and transparent research practices, students learn to build evidence-based marketing strategies that stand up to scrutiny from both C-suite executives and academic researchers.

Key topics include:

- **Consumer behavior in digital environments**: Applying marketing and consumer behavior methods to understand technology-mediated purchasing decisions
- **Psychological foundations of marketing technology**: How cognitive and emotional processes influence consumer responses to AI, personalization, and digital experiences
- **Human-computer interaction for marketers**: Designing user experiences that drive engagement, conversion, and brand loyalty
- **Data science for behavioral insights**: Using advanced analytics to uncover patterns in consumer behavior and optimize marketing strategies
- **Open science and reproducible research**: Implementing transparent research practices using tools like OSF (Open Science Framework) and preregistration
- **Ethics and privacy**: Navigating the responsible use of technology in marketing

The centerpiece of the course is an empirical group project where student teams conduct original behavioral research addressing real marketing challenges. Teams will integrate perspectives from all four disciplines—marketing, psychology, human-computer interaction, and data science—to develop comprehensive solutions. From developing research questions and preregistering hypotheses to collecting data and presenting findings, this project provides experience in the full research cycle with direct application to marketing practice.

Throughout the course, students will engage with both **academic experts** and **industry practitioners** through guest lectures, gaining unique insights into how behavioral science drives innovation in leading companies. Using professional tools including **R for statistical analysis**, **Qualtrics for experimental design**, and **OSF for transparent research**, students will develop skills directly applicable to evidence-based marketing careers.

Students will have an insight into the **Institute of Behavioral Science and Technology (IBT-HSG)**, positioning themselves at the forefront of the digital transformation.

Structure and indications of the learning and teaching design

This course employs an innovative **hybrid learning format** combining:

1. **Interactive kick-off session**: Setting the foundation for understanding behavioral science in marketing contexts
2. **Research immersion**: Students present and critically analyze research papers from marketing, psychology, HCI, and data science journals, developing their ability to synthesize interdisciplinary insights for marketing practice
3. **Intensive bootcamps**: Two full-day hands-on workshops where students:
    - Master observational research methods from consumer research and open science practices including preregistration
    - Apply psychological theories to design effective A/B tests and marketing experiments
    - Develop data science skills using R for behavioral analysis
    - Explore ethical considerations in behavioral marketing from all four disciplinary perspectives
4. **Empirical group project**: This comprehensive research experience challenges teams to:
    - Identify pressing marketing problems where behavioral insights can make a difference

- Develop theoretically grounded hypotheses drawing from marketing, psychology, HCI, and data science literature
- Design studies that integrate consumer research methods, psychological experiments, user testing, and data analytics
- Preregister their study design following open science principles
- Collect and analyze real behavioral data using advanced statistical techniques
- Translate findings into actionable marketing recommendations
- Present results professionally, demonstrating both scientific rigor and business relevance

Throughout the project, teams receive personalized coaching sessions by the instructors.

5. **Professional presentation**: Final presentations showcase research findings in a professional setting, simulating real-world marketing presentations

The course design emphasizes **learning by doing**, with immediate application of concepts to marketing problems. By the end, students will have a complete research project in their portfolio, demonstrating their ability to:

- Identify and solve real marketing challenges using behavioral insights
- Design and execute rigorous experiments that yield actionable results
- Communicate complex findings to diverse stakeholders
- Apply ethical, evidence-based approaches to marketing innovation

This hands-on approach ensures graduates can immediately contribute to marketing teams in consulting firms, tech companies, consumer brands, or innovative startups.

Literature

Core readings will include:

- Recent publications from top-tier marketing and psychology journals
- Harvard Business Review articles on marketing innovation and technology
- Open science resources and methodological guides

Additional materials including industry reports, guest speaker recommendations, and real-world datasets may be provided via Canvas throughout the course.

Additional information

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Examination information

Examination sub part/s

1. Examination sub part (1/3)

Examination modalities

Examination type	Presentation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Group work group grade
Weighting	50%
Duration	--

Languages
Question language: English
Answer language: English

Remark
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Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids
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2. Examination sub part (2/3)

Examination modalities

Examination type	Presentation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	30%
Duration	--

Languages
Question language: English
Answer language: English

Remark
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Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids
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3. Examination sub part (3/3)

Examination modalities

Examination type	Active participation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Synchronous

Examination location	On Campus
Grading type	Individual work individual grade
Weighting	20%
Duration	--

Languages
Question language: English
Answer language: English

Remark
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Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids
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Content

Individual student presentation

In the individual presentations, students will present an original research article on behavioral science and technology. Emphasis will be placed on accurately and concisely presenting and critically reflecting the article.

Active participation

The participation and presence part of the examination will be evaluated based on the active participation during the sessions and the engagement in the Canvas online forum.

Group presentation and report

In the presentation and report part of the examination, groups of students will be required to apply their newly acquired practical method skills in an empirical project. Specifically, they will need to either conduct an empirical study or find a suitable dataset to apply the acquired analytical skills to and write a written report on the applied method and its results. At the end of the course, the groups will present their method and results to the class and answer questions from their peers and the lecturer team.

Presentation slides must be uploaded to Canvas **two days** before the presentation (PDF).

Timing of the presentation: 20 min + 10 min Q&A + general discussion at the end.

Literature

In addition to the lecture slides and the content of all in-class discussions, relevant journal articles, digital content and additional readings will be announced and made available during the course.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW51) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 43 (Freitag, 24. October 2025) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 34 (Thursday, 21 August 2025);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 42 (Monday, 13. October 2025);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 43 (Freitag, 24. October 2025);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 45 (Monday, 03. November 2025).